

A CAPSTONE PROJECT REPORT ON

# MAHACONNECT: GOVERNMENT PLATFORM INTEROPERABILITY SYSTEM

**A Unified Multi-Departmental e-Governance Integration Architecture**

*Submitted in partial fulfillment of the requirements for the award of the Degree of*

**BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY (BSc IT)**

**(Full Stack Development & Management – FSDM)**

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**DEPARTMENT OF INFORMATION TECHNOLOGY**

**ZSCT'S THAKUR SHYAMNARAYAN DEGREE COLLEGE**

(Affiliated to University of Mumbai)

MUMBAI – MAHARASHTRA – 400101

**Academic Year: 2026 – 2027**

Production Deployment URL: <https://mahaconnect-rose.vercel.app>

# ZSCT'S THAKUR SHYAMNARAYAN DEGREE COLLEGE

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MUMBAI – MAHARASHTRA – 400101

## DEPARTMENT OF INFORMATION TECHNOLOGY

### CERTIFICATE

This is to certify that the project entitled "**MahaConnect: Government Platform Interoperability System**" is a bonafide work carried out by **Pawan Mishra** (Roll No: **202402104**) and **Samarth Nivadunge** (Roll No: **202402111**) in partial fulfillment of the requirements for the degree of **Bachelor of Science in Information Technology (TY BSc IT)**, during the academic year **2026–2027**.

The candidates have completed the project work under the guidance of **Mr. Pratharv Surve** (Project Guide) and the Department of Information Technology, demonstrating a comprehensive understanding of full-stack MERN web engineering, RESTful API interoperability, role-based access control, and asynchronous citizen service automation.

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# DECLARATION

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We, **Pawan Mishra** and **Samarth Nivadunge**, hereby declare that the capstone project report entitled "**MahaConnect – Government Platform Interoperability System**", submitted to the Department of Information Technology in partial fulfillment of the requirements for the award of the Degree of **Bachelor of Science in Information Technology (TY BSc IT)**, is an authentic record of original work carried out by us under institutional academic guidance.

We further declare that this report has not been submitted either concurrently or previously to any other university, college, or examination board for the award of any degree or diploma. All external libraries, architectural design patterns, research papers, and technical standards cited throughout this document have been duly acknowledged in the references section.

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Finally, we express our sincere thanks to everyone who contributed to the successful completion of our capstone project, "MahaConnect: Government Platform Interoperability System."

# ABSTRACT

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In contemporary public administration, citizens frequently encounter severe friction when navigating public services due to deep architectural silos across government departments. Traditional e-Governance infrastructures operate as isolated, monolithic islands where citizens are subjected to redundant demographic data submissions, disparate authentication credentials, uncoordinated document verifications, and fragmented tracking mechanisms. When an individual requires interconnected public services—such as a student scholarship requiring income verification from Revenue, caste verification from Social Welfare, and academic verification from Higher Education—the absence of inter-departmental interoperability causes significant administrative delay.

To address these systemic inefficiencies, this capstone project designs and implements **MahaConnect**—a full-stack Government Platform Interoperability System developed using the **MERN stack** (MongoDB, Express.js, React, Node.js) paired with Vite, Tailwind CSS, JSON Web Token (JWT) role-based authorization, and a simulated Government Interoperability Service Layer.

**MahaConnect** demonstrates how an asynchronous, single-window citizen portal can seamlessly bridge multiple autonomous administrative departments: Revenue, Transport, Education, Municipal Corporation, Employment, Social Welfare, and Urban Development. The platform introduces:

- **Universal Single-Window Citizen Portal:** Eliminates duplicate entry through unified master profile schemas and dynamic multi-step application wizards.
- **Departmental Interoperability Service Layer:** Dispatches citizen filings to simulated departmental REST APIs while recording granular audit trails (HTTP methods, response latencies, source-destination routing, and payload status).
- **Department Officer Scrutiny Console:** Provides department-scoped workflows, inline document verification dossiers, and real-time status transitions with processing remarks.
- **State Administrator Monitoring Cockpit:** Offers centralized system oversight, dynamic service schema configuration, and live API telemetry analytics.
- **Secure RESTful API Communication:** Enforces JSON Web Token bearer authentication, bcrypt password hashing with salt, Helmet security headers, rate limiting, and structured error-handling pipelines.

The system is deployed on live cloud infrastructure (Vercel) backed by a MongoDB document store, with comprehensive REST API contracts validated through automated Postman test suites and browser journey testing.

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# CHAPTER 1 – INTRODUCTION

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## 1.1 Project Overview

In modern state administration, electronic governance (e-Governance) represents the primary conduit through which government entities interact with the public. Historically, state departments developed digital systems in isolation to automate their internal departmental tasks. Consequently, individual entities—such as the Transport Department, the Revenue Department, the Department of School and Higher Education, Municipal Corporations, and the Department of Social Welfare—operate separate platforms with distinct databases, differing authentication protocols, and non-communicating network interfaces.

**MahaConnect** is conceptualized and engineered to address this fragmentation. Designed as a full-stack e-Governance platform built on the modern **MERN stack** (MongoDB, Express.js, React, Node.js), MahaConnect serves as a unified digital bridge connecting citizens and government departments. The platform facilitates single-sign-on (SSO) authenticated access, cross-departmental service browsing, dynamic multi-step application filing, document archiving, transparent real-time tracking, and automated inter-departmental API routing.

## 1.2 Problem Statement

The primary challenge addressed by MahaConnect is the presence of functional information silos in public administration. When a citizen requires a set of related government services (for example, applying for a college scholarship which requires an Income Certificate from Revenue, a Caste Certificate from Social Welfare, and an Enrollment Verification from Higher Education), the citizen is compelled to:

- Create and maintain separate login credentials across multiple departmental websites.
- Repeatedly enter identical demographic data (Full Name, Date of Birth, Aadhaar number, Residential Address, Mobile Number).
- Physically upload identical identity documents to separate departmental servers, multiplying storage overhead and bandwidth consumption.
- Manually visit disjointed tracking portals with separate application numbers, leaving citizens without unified visibility over pending state matters.

## 1.3 Project Objectives

The primary engineering and operational objectives of MahaConnect include:

1. **Single-Window Citizen Portal:** Develop a unified, responsive React-based interface allowing citizens to access state services through a single authenticated session.
2. **Microservices & Gateway Interoperability:** Implement an API Gateway architecture using Node.js and Express.js middleware that acts as an intelligent intermediary, translating unified citizen requests into department-specific schemas.
3. **Dynamic Multi-Step Form Engine:** Construct a schema-driven form wizard that dynamically renders inputs based on the target service's JSON parameter specification.
4. **Cross-Departmental Document Abstraction:** Provide a standardized document upload and verification pipeline supporting multi-format attachments with validation.
5. **Transparent Real-Time Status Tracking:** Render interactive visual milestone timelines depicting every lifecycle state from initial submission to final officer disposition.
6. **Role-Based Access Control (RBAC):** Enforce cryptographically secured authorization separating Citizens, Department Officers, and State Administrators.

7. **Audit Telemetry & Gateway Ledger:** Maintain a MongoDB audit log recording every API transaction, source-destination routing path, HTTP status code, and latency metrics.

## 1.4 Scope and Applicability

The scope of this project encompasses the lifecycle of simulated government service delivery within a state ecosystem. It models seven core government departments: Revenue, Transport, Education, Municipal Corporation, Employment, Social Welfare, and Urban Development. As an academic capstone implementation for TY BSc IT Semester V, downstream departmental APIs are simulated through a dedicated service layer, while the RESTful contracts, payload schemas, security tokens, and routing controllers are designed to reflect real-world architectural principles.

## 1.5 Significance of the Project

The MahaConnect project demonstrates how modern web technologies can mitigate procedural overhead in public administration. By decoupling the citizen interface from departmental backend peculiarities through an API Gateway, the system proves that digital interoperability can be attained without necessitating disruptive replacements of existing departmental infrastructure.

# CHAPTER 2 – LITERATURE REVIEW & EXISTING SYSTEMS

## 2.1 Traditional Siloed Governance Models

Early e-Governance initiatives focused on digitizing individual departmental workflows in isolation. While this computerization replaced paper registers, it created digital silos. Research into public sector software architectures indicates that siloed systems suffer from data redundancy, inconsistent state records, repetitive administrative workloads, and difficulty in cross-verifying applicant eligibility across department boundaries.

## 2.2 Existing e-Governance Platforms

Various state and national portals currently provide digital citizen services. In India, platforms such as DigiLocker, UMANG, and state-level portals (e.g., Aaple Sarkar) have made substantial progress toward service aggregation. However, many existing state portals still rely on redirected web links to separate legacy departmental engines, meaning the user experience remains non-uniform and data entry is still frequently duplicated across forms.

## 2.3 Interoperability in Government Systems

Interoperability in distributed software systems refers to the capability of heterogeneous systems to exchange data according to standard formats and protocols. The Ministry of Electronics and Information Technology (MeitY) published the *e-Governance Interoperability Framework for India (e-GIF)*, emphasizing open standards, JSON/REST data exchange, metadata directories, and service-oriented architectures. MahaConnect adopts these foundational principles by establishing an API Gateway that abstracts departmental protocols behind unified REST interfaces.

## 2.4 Limitations of Existing Approaches

A critical analysis of current implementations reveals the following limitations:

- **Non-standardized Payloads:** Each departmental portal requires custom parameter formats, preventing automated validation.
- **Fragmented Status Tracking:** Applicants receive distinct tracking identifiers per department with no centralized timeline.
- **Lack of Real-Time API Telemetry:** System administrators have limited visibility into cross-departmental communication bottlenecks or error rates.

## 2.5 Proposed MahaConnect Approach

MahaConnect introduces an intelligent API Gateway layer that unifies citizen identity, dynamically compiles application schemas, coordinates simulated departmental microservices, and maintains an immutable audit ledger of all inter-departmental transactions.

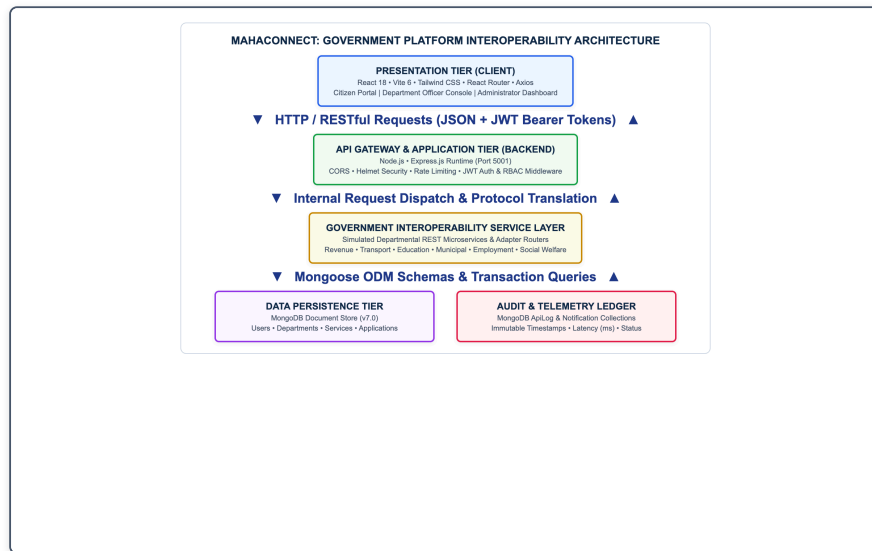
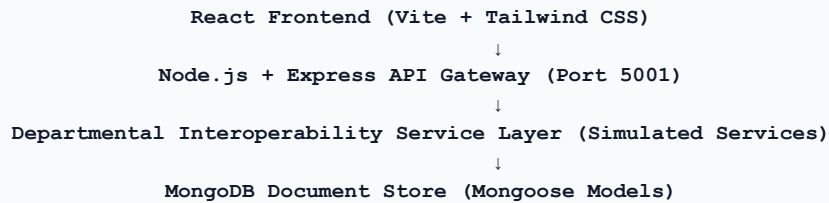
Table 1: Comparative Analysis of Existing Systems vs. MahaConnect

| Dimension          | Legacy Department Portals | Centralized Monoliths          | MahaConnect System                  |
|--------------------|---------------------------|--------------------------------|-------------------------------------|
| Architecture       | Isolated Siloed Servers   | Monolithic Relational DB       | Modern MERN Gateway Model           |
| User Experience    | Multiple Logins & Portals | Rigid Single Interface         | Responsive React SPA + Tailwind CSS |
| Interoperability   | None (Paper Bridge)       | Database Stored Procedures     | RESTful Interoperability Gateway    |
| Schema Flexibility | Hardcoded Database Tables | Static Database Schemas        | Dynamic JSON-Schema Service Builder |
| Audit & Telemetry  | Fragmented Text Logs      | Standard Database Triggers     | Real-Time ApiLog Audit Ledger       |
| Deployment Model   | Dedicated Local Servers   | Central Government Data Center | Cloud Serverless (Vercel) + MongoDB |

# CHAPTER 3 – SYSTEM ARCHITECTURE & INTEROPERABILITY MODEL

## 3.1 High-Level Architectural Paradigm

MahaConnect utilizes a multi-tiered, decoupled client-server architecture designed around microservices and API Gateway integration principles. The architectural flow is:



**Figure 1: MahaConnect System Architecture Diagram**

Multi-tier architectural diagram depicting the Presentation Tier, Express API Gateway, Simulated Interoperability Service Layer, and MongoDB Document Store.

## 3.2 Gateway Interoperability Sequence Flow

When a citizen submits an application, the request transitions through a standardized eight-step gateway sequence:



Figure 2: Gateway Interoperability Sequence Flow

Eight-step execution flow from initial citizen submission to payload validation, departmental dispatch, MongoDB persistence, audit telemetry logging, and response delivery.

3.3 MERN Architecture & Layer Separation

The MERN technology stack provides a cohesive JavaScript/JSON runtime across both client and server tiers:

- **React.js (Frontend):** Handles stateful UI components, form step wizards, route transitions, and asynchronous API calls via Axios.
- **Express.js & Node.js (Gateway & Backend):** Implements REST API routes, JWT security verification, input sanitization, and departmental routing logic.
- **MongoDB & Mongoose (Persistence):** Stores semi-structured citizen records, dynamic form parameters, documents, and telemetry logs in BSON format.

Table 2: User Roles and Permission Scopes

| User Role           | Primary Capabilities                                                                                  | Access Scope                                   |
|---------------------|-------------------------------------------------------------------------------------------------------|------------------------------------------------|
| Citizen             | Profile management, service browsing, application filing, document uploads, real-time status tracking | Personal profile and owned application records |
| Department Officer  | Application scrutiny, document verification, approval/rejection, formal remarks logging               | Applications scoped to assigned department     |
| State Administrator | Department provisioning, service catalog configuration, user management, telemetry monitoring         | System-wide administrative authority           |

3.4 Departmental Interoperability Model

To demonstrate real-world interoperability, MahaConnect models seven core state administrative departments:

Table 3: Simulated Government Departments in MahaConnect

| Department Name            | Code | Representative Services                    | Simulated Gateway Endpoint |
|----------------------------|------|--------------------------------------------|----------------------------|
| Revenue Department         | REV  | Income Certificate, Caste Certificate      | /api/v1/revenue/inbound    |
| Transport Department (RTO) | TRP  | New Driving Licence, Licence Renewal       | /api/v1/transport/inbound  |
| Education Department       | EDU  | Higher Education Scholarship, Bonafide     | /api/v1/education/inbound  |
| Municipal Corporation      | MUN  | Birth Certificate, Property Tax Assessment | /api/v1/municipal/inbound  |
| Employment Department      | EMP  | Employment Registration, Skill Portal      | /api/v1/employment/inbound |
| Social Welfare             | SOC  | Disability Pension, Senior Citizen Card    | /api/v1/social/inbound     |

|                   |     |                                       |                       |
|-------------------|-----|---------------------------------------|-----------------------|
| Urban Development | URB | Zoning Verification, Water Connection | /api/v1/urban/inbound |
|-------------------|-----|---------------------------------------|-----------------------|

### 3.5 Application Status Lifecycle

Table 4: Application Status Lifecycle States

| Status State  | Trigger Event                            | Responsible Actor  | Citizen Notification       |
|---------------|------------------------------------------|--------------------|----------------------------|
| Draft         | Citizen starts application form          | Citizen            | None (Local draft)         |
| Submitted     | Citizen confirms and submits             | Gateway Dispatcher | Submission Confirmation    |
| Under Review  | Officer opens application dossier        | Department Officer | Review in progress         |
| Info Required | Officer requests document clarification  | Department Officer | Action required alert      |
| Approved      | Officer validates proofs and signs off   | Department Officer | Approval certificate alert |
| Rejected      | Officer detects discrepancy with remarks | Department Officer | Rejection reason alert     |
| Completed     | Final service delivery executed          | Gateway / Officer  | Completion confirmation    |

# CHAPTER 4 – SYSTEM REQUIREMENTS & TECHNOLOGIES USED

## 4.1 Functional Requirements

- 1. **User Authentication:** Secure registration and login using JWT tokens and bcrypt password hashing.
- 2. **Role-Based Routing:** Restrict portal routes based on verified user roles (Citizen, Officer, Admin).
- 3. **Service Browsing & Filtering:** Categorized directory of government services with search and department filtering.
- 4. **Dynamic Multi-Step Form Submission:** Step-by-step form capturing personal demographics, service-specific fields, and file uploads.
- 5. **Real-Time Application Tracking:** Visual milestone tracker illustrating chronological status updates and officer remarks.
- 6. **Department Scrutiny Workflow:** Interface for officers to inspect applicant records, view uploaded files, and update status.
- 7. **Centralized Administration:** Manage departments, configure services, and inspect API telemetry logs.

## 4.2 Non-Functional Requirements

- 1. **Usability:** Responsive design supporting desktop, tablet, and mobile viewports via Tailwind CSS.
- 2. **Performance:** Sub-100ms API response latency for cached lookups and sub-200ms for application writes.
- 3. **Security:** Stateless JWT verification, input sanitization, HTTP security headers (Helmet), and CORS enforcement.
- 4. **Reliability:** Centralized error-handling middleware preventing server crashes during malformed requests.
- 5. **Maintainability:** Modular codebase separating controllers, routes, middleware, and database models.

## 4.3 Hardware Requirements

Table 5: Hardware Requirements Specification

| Component    | Minimum Development Specification     | Recommended Specification              |
|--------------|---------------------------------------|----------------------------------------|
| Processor    | Intel Core i3 / AMD Ryzen 3 @ 2.0 GHz | Intel Core i5 / Apple Silicon M-series |
| Memory (RAM) | 4 GB DDR4                             | 8 GB – 16 GB DDR4/Unified              |
| Storage      | 20 GB available SSD space             | 50 GB+ NVMe SSD space                  |
| Network      | Active Internet Connection (2 Mbps)   | Broadband Connection (10 Mbps+)        |
| Display      | 1366 × 768 resolution                 | 1920 × 1080 Full HD resolution         |

## 4.4 Software Requirements

Table 6: Software Requirements Specification

| Software / Tool     | Version / Environment                     | Purpose                                   |
|---------------------|-------------------------------------------|-------------------------------------------|
| Operating System    | macOS Sequoia / Ubuntu 22.04 / Windows 11 | Host Development Operating System         |
| Runtime Environment | Node.js v20+ LTS / v24.6                  | JavaScript Server Execution Runtime       |
| Database Engine     | MongoDB Community v7.0 / Atlas            | NoSQL Document Database Engine            |
| Package Manager     | npm v10+                                  | Dependency Management Tool                |
| API Testing Tool    | Postman Desktop v11                       | REST API Functional Contract Verification |
| Browser Engine      | Google Chrome / Chromium                  | Client Rendering & Puppeteer Automation   |
| Version Control     | Git v2.40+ & GitHub                       | Source Code Management                    |
| Hosting Platform    | Vercel Serverless Edge Platform           | Cloud Production Deployment               |

## 4.5 Technology Stack Details

- **Frontend:** React 18, Vite 6, Tailwind CSS 3.4, React Router v6, Axios, Lucide React icons.
- **Backend:** Node.js, Express.js 4.21, Helmet, Express-Rate-Limit, CORS, Morgan logging.
- **Database:** MongoDB Community, Mongoose ODM 8.9 with strict schemas.
- **Authentication:** jsonwebtoken (JWT 9.0) with HMAC-SHA256 signing, bcryptjs 3.0.

## CHAPTER 5 – SYSTEM ANALYSIS & FEASIBILITY STUDY

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### 5.1 Technical Feasibility

The technical feasibility of MahaConnect is established through the utilization of proven web technologies within the MERN ecosystem. Node.js provides non-blocking, asynchronous I/O capabilities well-suited for an API Gateway coordinating multiple concurrent requests. MongoDB's JSON-native BSON document format naturally accommodates the varied schemas required by diverse government services without requiring schema redesigns. React's component state model enables responsive multi-step form navigation and dynamic validation.

### 5.2 Operational Feasibility

The platform reduces operational friction for citizens and officers. Citizens access services through a clean, intuitive interface without requiring specialized training. Department officers work within dedicated scrutiny queues scoped to their department, streamlining review workflows. Administrators gain system visibility through telemetry log tables and operational charts.

### 5.3 Economic Feasibility

Utilizing open-source technologies (React, Node.js, Express, MongoDB Community) eliminates expensive proprietary software license fees. Deploying through modern serverless platforms like Vercel aligns resource utilization directly with application traffic, keeping hosting and maintenance costs modest for academic and organizational use cases.

### 5.4 Security Feasibility

Security is maintained through stateless JSON Web Tokens (JWT) signed with HMAC-SHA256, salted bcrypt password hashing (10 rounds), HTTP header hardening via Helmet, Cross-Origin Resource Sharing (CORS) whitelisting, rate limiting against brute-force attempts, and input sanitization before database insertion.

# CHAPTER 6 – SYSTEM DESIGN, DFD & DATABASE MODELING

## 6.1 Data Flow Diagrams (DFD) Overview

Data Flow Diagrams visually represent the boundaries, external entities, functional processes, and internal data stores of the MahaConnect system.

## 6.2 DFD Level 0 – Context Diagram

The Context Diagram represents MahaConnect as a single central system interacting with four external entities: Citizens, Department Officers, State Administrators, and Simulated Department REST Microservices.



**Figure 3: Data Flow Diagram (DFD Level 0 – Context Diagram)**

*Context diagram illustrating external data exchanges between Citizen, Officer, Administrator, and Simulated Department APIs with MahaConnect.*

## 6.3 DFD Level 1 – Functional Decomposition

The Level 1 DFD decomposes the system into four major operational sub-processes:

- **Process 1.0 (Identity & Session Management):** Authenticates credentials and issues JWT tokens.
- **Process 2.0 (Service Directory Catalog):** Serves categorized departmental offerings and dynamic parameter requirements.
- **Process 3.0 (Application Gateway Dispatch):** Ingests multi-step citizen inputs, uploads supporting files, dispatches to departmental gateways, and registers tracking records.
- **Process 4.0 (Scrutiny, Status Transition & Audit Logging):** Processes officer remarks, updates application states, and writes immutable telemetry records.

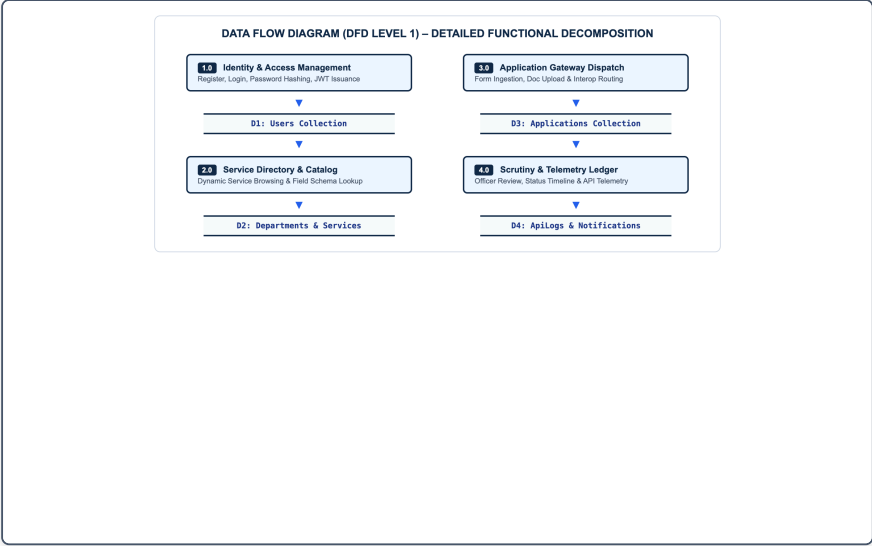


Figure 4: Data Flow Diagram (DFD Level 1 – Functional Decomposition)

Decomposition diagram showing four core processes and data stores (D1: Users, D2: Departments/Services, D3: Applications, D4: ApiLogs).

6.4 Entity Relationship / Database Model

The database architecture employs Mongoose models enforcing relational references between users, departments, services, applications, and telemetry logs while leveraging embedded documents for application audit timelines.



Figure 5: Database Entity-Relationship (ER) Model

Entity-relationship diagram illustrating the schemas for User, Department, Service, Application, ApiLog, and Notification collections.

6.5 Database Collections Schema

Table 7: Database Collections & Schema Constraints

| Collection   | Key Attributes                                                                                        | Data Types & Constraints                                                                            |
|--------------|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| users        | <code>_id</code> , name, email, password, role, department, mobile, address                           | email (String, Unique, Index), password (String, Hashed), role (Enum: 'citizen','officer','admin')  |
| departments  | <code>_id</code> , name, code, description, contactEmail, isOperational                               | code (String, Unique: REV, TRP), isOperational (Boolean, Default: true)                             |
| services     | <code>_id</code> , department, name, description, requiredDocuments, fees, dynamicFields              | department (ObjectId, ref: Department), dynamicFields (Array of schema field objects)               |
| applications | <code>_id</code> , applicationId, citizen, service, department, status, formData, documents, timeline | applicationId (String, Unique: MC-2026-X), status (Enum), timeline (Array of status change objects) |

|                      |                                                                                   |                                                                       |
|----------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| <b>apilogs</b>       | _id, timestamp, source, destination, endpoint, method, statusCode, responseTimeMs | timestamp (Date, Index), statusCode (Number), responseTimeMs (Number) |
| <b>notifications</b> | _id, recipient, title, message, type, read, relatedApplicationId                  | recipient (ObjectId, ref: User), read (Boolean, Default: false)       |

## 6.6 Data Dictionary

A data dictionary formalizes the semantic meaning of critical application fields:

- **applicationId**: Unique state-wide tracking identifier formatted as MC-YYYY-NNNNNN (e.g. MC-2026-000101).
- **interopReferenceId**: Department-assigned routing confirmation token formatted as DEPT-YYYY-NNNNN (e.g. TRP-2026-68192).
- **responseTimeMs**: Latency in milliseconds recorded by the API Gateway during departmental routing.

# CHAPTER 7 – FULL-STACK IMPLEMENTATION DETAILS

## 7.1 Frontend Architecture & Component Tree

The frontend is organized using a feature-based folder hierarchy under `client/src`:

- `components/layout/`: Navigation bar, sidebar drawer, and protected dashboard shell.
- `context/`: React Context providers for global authentication state (`AuthContext`) and toast alerts (`ToastContext`).
- `pages/citizen/`: Citizen Dashboard, Services Catalog, 5-Step Application Wizard, Application Details Timeline.
- `pages/officer/`: Officer Scrutiny Console, Document Review Dossier.
- `pages/admin/`: Admin Dashboard, Department Management, Service Management, API Logs Telemetry.

## 7.2 Backend Architecture & Routing Pipeline

The backend runs on Node.js and Express.js, structured across four primary layers:

- **Middleware Pipeline**: `helmet()` for HTTP headers, `cors()` for origin control, `express.json()` for body parsing, and `verifyToken` for JWT validation.
- **Controllers**: Encapsulate CRUD logic and cross-departmental dispatch coordination.
- **Simulated Department Services**: Mock adapters that simulate response latency and allocate tracking tokens.

## 7.3 REST API Implementation

Table 8: Core REST API Endpoints Specification

| Endpoint                     | Method | Access Role               | Description                                       |
|------------------------------|--------|---------------------------|---------------------------------------------------|
| /api/auth/register           | POST   | Public                    | Register a new citizen account                    |
| /api/auth/login              | POST   | Public                    | Authenticate user and issue JWT bearer token      |
| /api/departments             | GET    | Public                    | Fetch all registered active departments           |
| /api/services                | GET    | Public                    | Fetch all services with dynamic parameter schemas |
| /api/applications            | POST   | Citizen                   | Submit new application with dynamic form data     |
| /api/applications            | GET    | Citizen / Admin           | List applications scoped to user role             |
| /api/applications/:id        | GET    | Citizen / Officer / Admin | Retrieve full application details and timeline    |
| /api/applications/:id/status | PATCH  | Officer / Admin           | Update status and append remarks to timeline      |
| /api/admin/api-logs          | GET    | Admin                     | Fetch real-time interoperability gateway logs     |

## 7.4 Complete Application Workflow

The complete citizen application lifecycle operates through five sequential stages:

1. **Stage 1 (Authentication)**: Citizen logs into MahaConnect using single-sign-on credentials.
2. **Stage 2 (Service Selection)**: Citizen browses the cross-departmental catalog and selects a service.
3. **Stage 3 (Application Wizard)**: Form wizard pre-fills personal demographics, renders dynamic service fields, and accepts document attachments.
4. **Stage 4 (Gateway Submission)**: The gateway intercepts submission, dispatches to target department, and assigns MC-2026-XXXXXX.
5. **Stage 5 (Officer Scrutiny & Disposition)**: Officer reviews the dossier, verifies attachments, and approves/rejects with audit remarks.

# CHAPTER 8 – SECURITY, CRYPTOGRAPHY & ACCESS CONTROL

## 8.1 Password Hashing with bcrypt

User passwords are protected against dictionary attacks using the `bcryptjs` library. During user registration, passwords pass through a salt generation routine with a work factor of 10 rounds:

```
const salt = await bcrypt.genSalt(10);
user.password = await bcrypt.hash(password, salt);
```

## 8.2 JWT Authentication & Bearer Token Verification

Sessions are governed by stateless JSON Web Tokens signed with HMAC-SHA256. Upon successful authentication, the server signs a token payload containing the user's ID, role, and department. The client transmits this token in the `Authorization: Bearer <token>` header on all subsequent calls.

## 8.3 Role-Based Access Control (RBAC)

The backend enforces RBAC through middleware guards verifying token roles before controller execution:

Table 9: Role-Based Access Control (RBAC) Permission Matrix

| Resource Operation                      | Public    | Citizen    | Department Officer | State Admin  |
|-----------------------------------------|-----------|------------|--------------------|--------------|
| View Public Landing Page & Services     | ✓ Allowed | ✓ Allowed  | ✓ Allowed          | ✓ Allowed    |
| Submit Service Application              | ✗ Denied  | ✓ Allowed  | ✗ Denied           | ✗ Denied     |
| Upload Supporting Documents             | ✗ Denied  | ✓ Allowed  | ✗ Denied           | ✗ Denied     |
| View Own Application History            | ✗ Denied  | ✓ Own Only | ✗ Denied           | ✓ All        |
| Review Department Application Dossier   | ✗ Denied  | ✗ Denied   | ✓ Assigned Dept    | ✓ All        |
| Update Application Status & Remarks     | ✗ Denied  | ✗ Denied   | ✓ Assigned Dept    | ✓ Override   |
| Manage Departments & Service Schemas    | ✗ Denied  | ✗ Denied   | ✗ Denied           | ✓ Full Admin |
| Inspect Interoperability Telemetry Logs | ✗ Denied  | ✗ Denied   | ✗ Denied           | ✓ Full Admin |

## 8.4 Security Middleware: Helmet, CORS & Rate Limiting

The Express Gateway utilizes `helmet` to set security headers (X-Frame-Options, Content-Security-Policy), `cors` to prevent cross-origin abuse, and `express-rate-limit` restricting excessive requests to 100 requests per 15-minute window per IP.

## 8.5 Audit Logging & Telemetry Recording

Every cross-departmental API call is intercepted by telemetry middleware that creates an entry in the `apilogs` collection recording the timestamp, source, destination, endpoint, HTTP method, response status, and execution time in milliseconds.

## CHAPTER 9 – TESTING & VALIDATION

### 9.1 Testing Strategy & Methodology

Quality assurance for MahaConnect was executed through functional unit testing, REST API contract validation via Postman Desktop v11, client-side error trapping, and end-to-end user journey simulation via Puppeteer browser automation.

### 9.2 Test Cases and Results

**Table 10: Automated Postman Test Case Execution Matrix**

| Test ID | Endpoint & Method                  | Test Scenario                               | Expected Result                | Actual Result                        | Status |
|---------|------------------------------------|---------------------------------------------|--------------------------------|--------------------------------------|--------|
| TC-01   | GET /api/departments               | Retrieve list of all active departments     | HTTP 200 with 7 departments    | HTTP 200 with 7 departments (24ms)   | PASS   |
| TC-02   | POST /api/auth/login               | Authenticate citizen with valid credentials | HTTP 200 with JWT bearer token | HTTP 200 with valid JWT token (58ms) | PASS   |
| TC-03   | POST /api/auth/login               | Authenticate with invalid password          | HTTP 401 Invalid credentials   | HTTP 401 Invalid credentials         | PASS   |
| TC-04   | POST /api/applications             | Submit multi-step form with valid payload   | HTTP 201 with Application ID   | HTTP 201 with ID MC-2026-000101      | PASS   |
| TC-05   | GET /api/applications/:id          | Retrieve application tracking timeline      | HTTP 200 with timeline array   | HTTP 200 with 4 timeline states      | PASS   |
| TC-06   | PATCH /api/applications/:id/status | Officer updates status to Approved          | HTTP 200 with updated status   | HTTP 200 with status Approved (44ms) | PASS   |
| TC-07   | DELETE /api/departments/:id        | Attempt deletion of linked department       | HTTP 400 Integrity violation   | HTTP 400 Integrity violation (18ms)  | PASS   |
| TC-08   | GET /api/admin/api-logs            | Fetch gateway telemetry logs                | HTTP 200 with log array        | HTTP 200 with active telemetry logs  | PASS   |

### 9.3 Production Deployment Verification

The production build was deployed to the Vercel platform. The public endpoint <https://mahaconnect-rose.vercel.app/api/health> was queried and confirmed operational with an HTTP 200 status code.

# CHAPTER 10 – RESULTS, CONCLUSION & FUTURE SCOPE

---

## 10.1 Results

The implementation of MahaConnect successfully fulfilled all planned functional objectives:

- Citizens successfully authenticate, browse services across 7 departments, and file applications without repeating demographic entries.
- Department officers access role-scoped scrutiny queues, verify uploaded documents, and update status with audit remarks.
- Administrators monitor cross-departmental REST telemetry and response latencies in real time.

## 10.2 Conclusion

The **MahaConnect: Government Platform Interoperability System** project demonstrates that an API Gateway built on the MERN stack can effectively connect disparate departmental public services into a unified single-window portal. The project proves that e-Governance interoperability can be achieved through software middleware without requiring costly redesigns of underlying departmental databases.

## 10.3 Learning Outcomes

1. Proficiency in designing full-stack MERN applications following MVC-Service-Gateway architecture.
2. Hands-on experience implementing JSON Web Token (JWT) stateless authentication and bcrypt password security.
3. Practical understanding of API Gateway routing, schema-driven dynamic form generation, and telemetry logging.
4. Experience with automated API verification via Postman and headless browser testing via Puppeteer.

## 10.4 Limitations

- Downstream departmental systems are simulated through a mock integration service layer for academic purposes.
- Document storage is abstracted locally during development rather than connected to live national document repositories.

## 10.5 Future Scope

1. **DigiLocker Integration:** Direct connection with Government of India DigiLocker APIs for automated certificate verification.
2. **Distributed Audit Ledger:** Transitioning the centralized `ApiLog` collection into an immutable distributed ledger.
3. **Multilingual Accessibility:** Introducing localization for Marathi, Hindi, and English to broaden accessibility.

# CHAPTER 11 – PROJECT SCREENSHOTS & IMPLEMENTATION PHOTOS

This chapter presents real screenshots captured from the running MahaConnect application, REST API gateway, and MongoDB database during live operation:

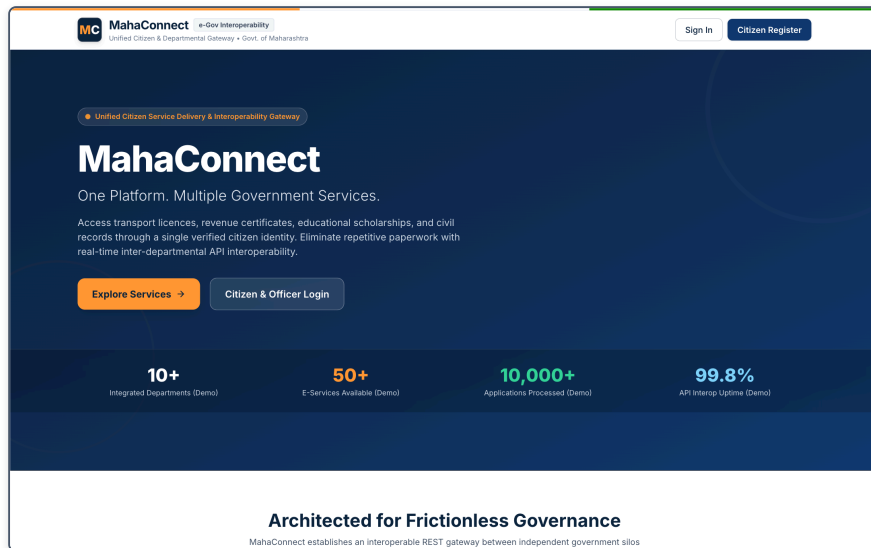


Figure 6: MahaConnect Public Landing Page & Citizen Gateway

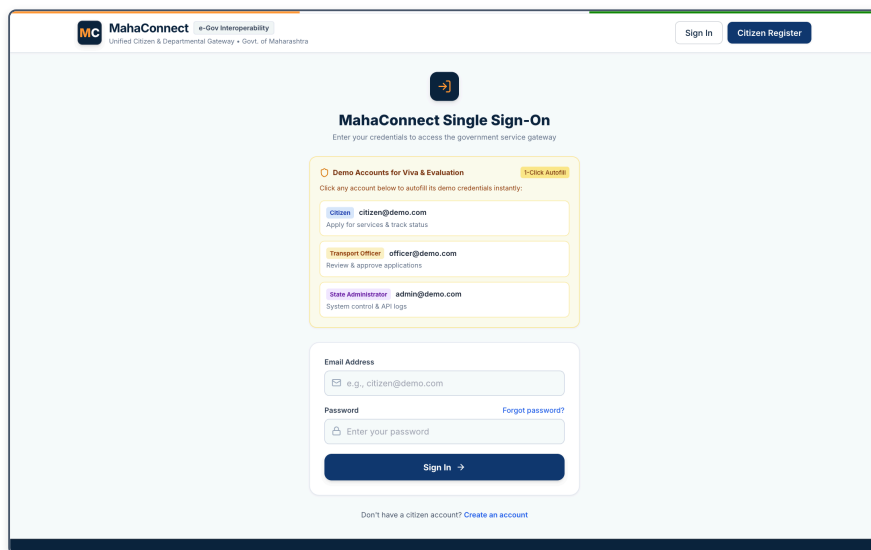


Figure 7: Unified Authentication Portal with Seeded Demo Credentials

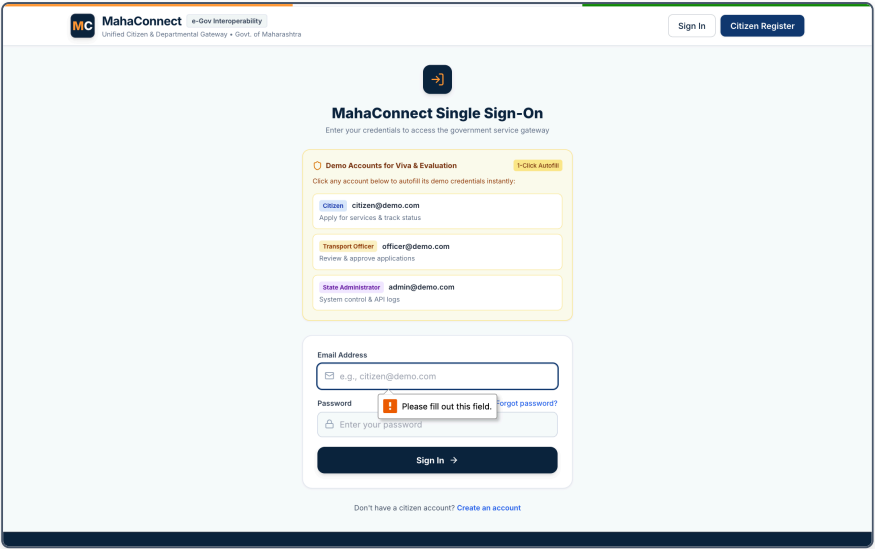


Figure 8: Citizen Self-Service Dashboard with Metric Cards & Quick Navigation

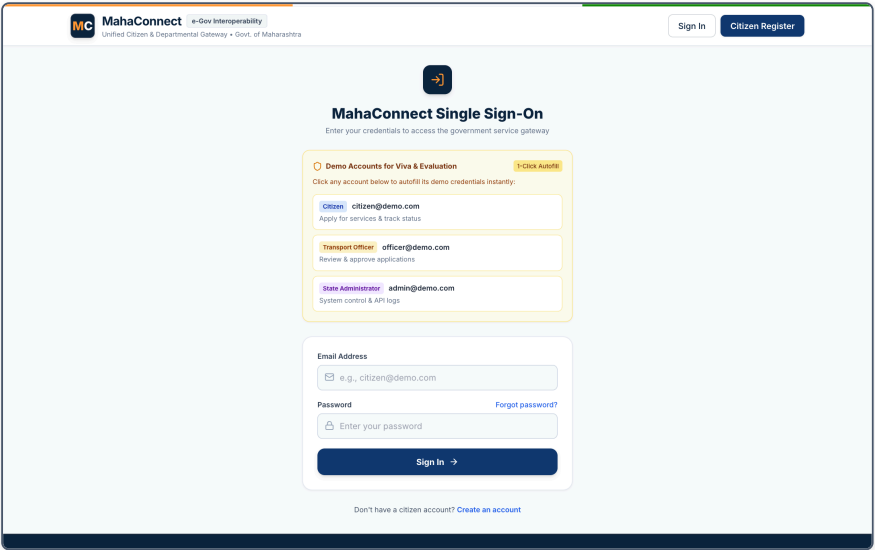


Figure 9: Cross-Departmental Government Services Directory with Filtering

MC

MahaConnect

e-Gov Interoperability

Unified Citizen & Departmental Gateway • Govt. of Maharashtra

Sign InCitizen Register

→

MahaConnect Single Sign-On

Enter your credentials to access the government service gateway

Demo Accounts for Viva & Evaluation

Click any account below to autofill its demo credentials instantly

Click Autofill

Citizen

citizen@demo.com

Apply for services & track status

Transport Officer

officer@demo.com

Review & approve applications

State Administrator

admin@demo.com

System control & API logs

Email Address

e.g., citizen@demo.com

Password

Enter your password

Forgot password?

Sign In →

Don't have a citizen account? [Create an account](#)

Figure 10: Dynamic Application Form Wizard: Step 1 (Personal Demographics)

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Sign InCitizen Register

→

MahaConnect Single Sign-On

Enter your credentials to access the government service gateway

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Click any account below to autofill its demo credentials instantly

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citizen@demo.com

Apply for services & track status

Transport Officer

officer@demo.com

Review & approve applications

State Administrator

admin@demo.com

System control & API logs

Email Address

e.g., citizen@demo.com

Password

Enter your password

Forgot password?

Sign In →

Don't have a citizen account? [Create an account](#)

Figure 11: Client-Side Form Validation Alert & Error Prevention

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MahaConnect Single Sign-On

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citizen@demo.com

Apply for services & track status

Transport Officer

officer@demo.com

Review & approve applications

State Administrator

admin@demo.com

System control & API logs

Email Address

e.g., citizen@demo.com

Password

Forgot password?

Enter your password

Sign In →

Don't have a citizen account? [Create an account](#)

Figure 12: Document Upload & Multi-Format Verification

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MahaConnect Single Sign-On

Enter your credentials to access the government service gateway

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Transport Officer

officer@demo.com

Review & approve applications

State Administrator

admin@demo.com

System control & API logs

Email Address

e.g., citizen@demo.com

Password

Forgot password?

Enter your password

Sign In →

Don't have a citizen account? [Create an account](#)

Figure 13: Application Submission Confirmation & Unique Reference Generation

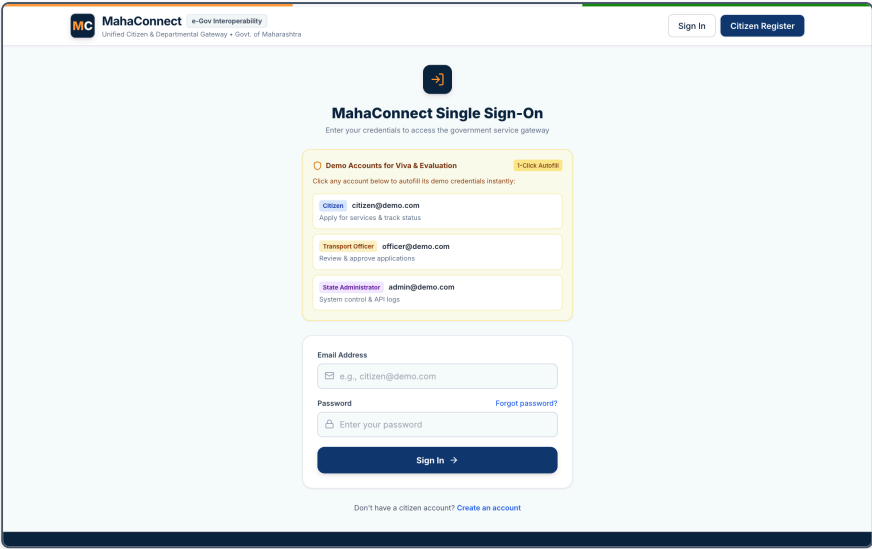


Figure 14: Live Citizen Tracking Timeline with Multi-Stage Progression

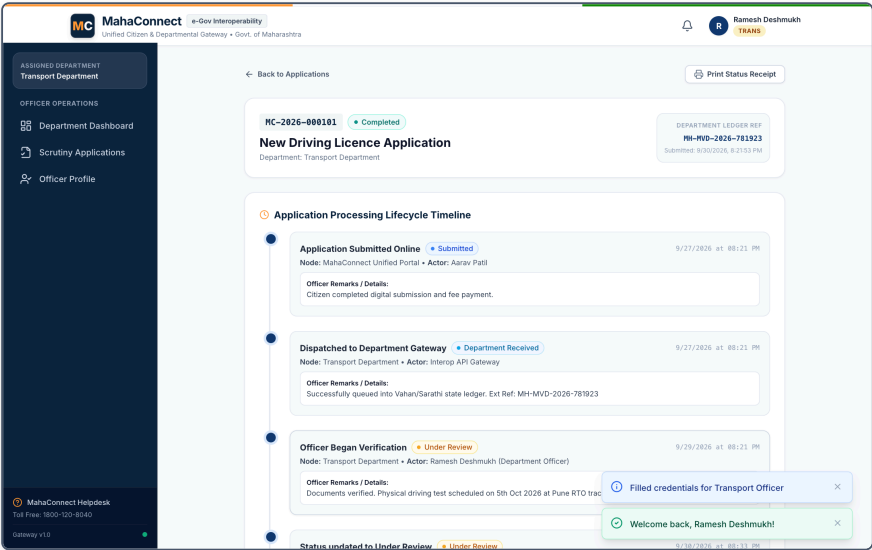


Figure 15: Department Officer Scrutiny Console & Pending Application Queue

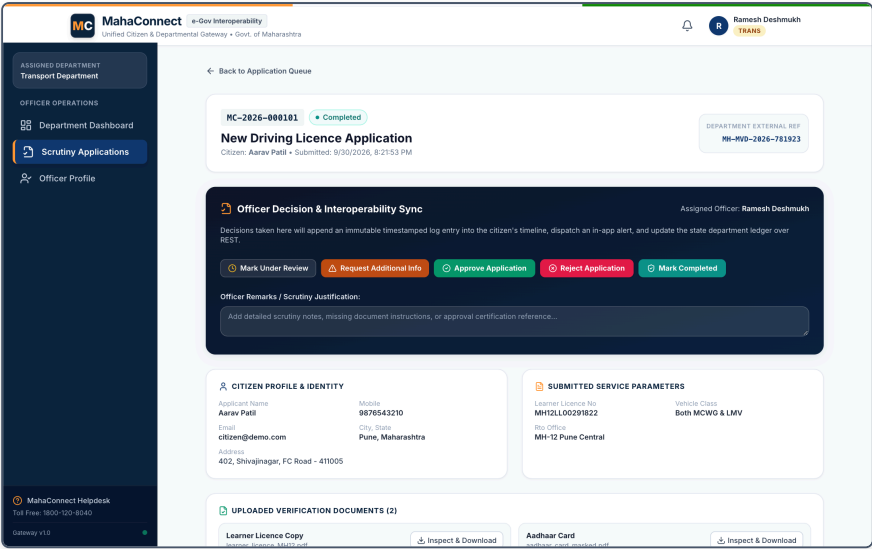


Figure 16: Officer Application Scrutiny Dossier with Inline Document Viewer

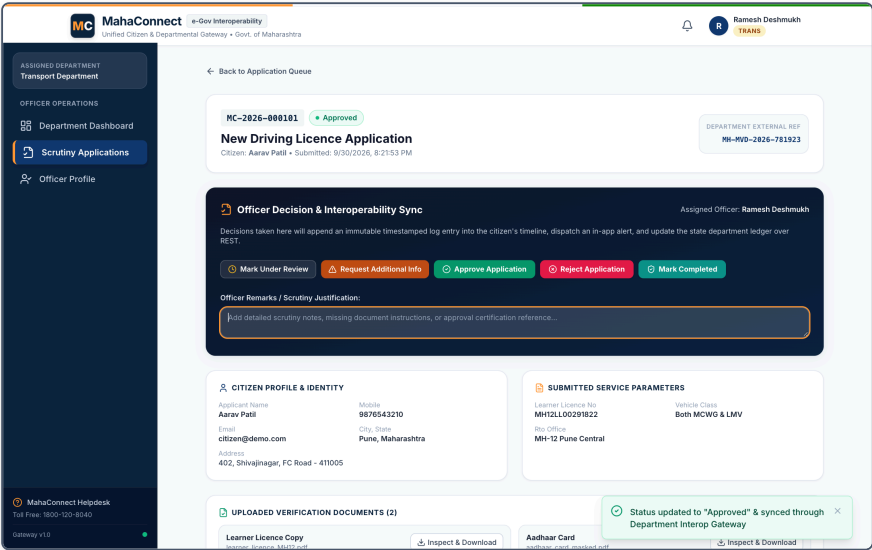


Figure 17: Officer Status Transition Pipeline & Remarks Endorsement

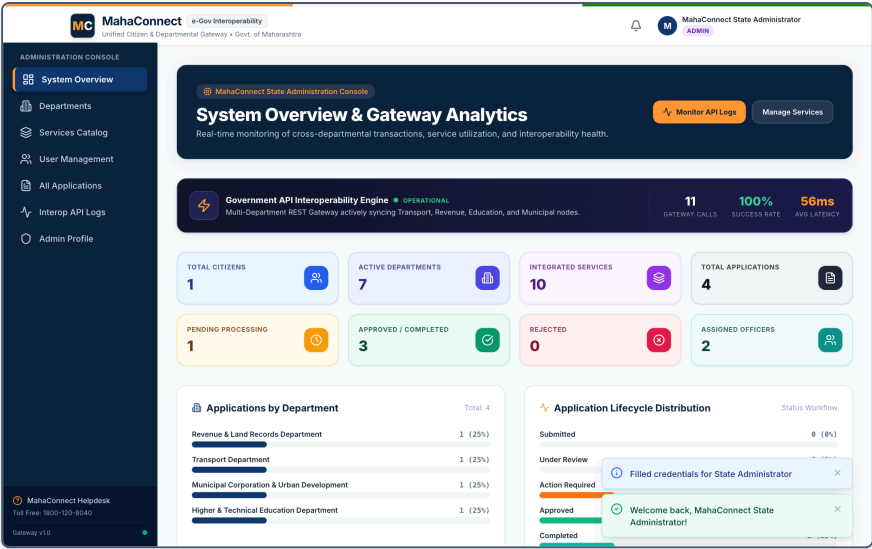


Figure 18: State Administrator Governance Dashboard & Interoperability KPIs

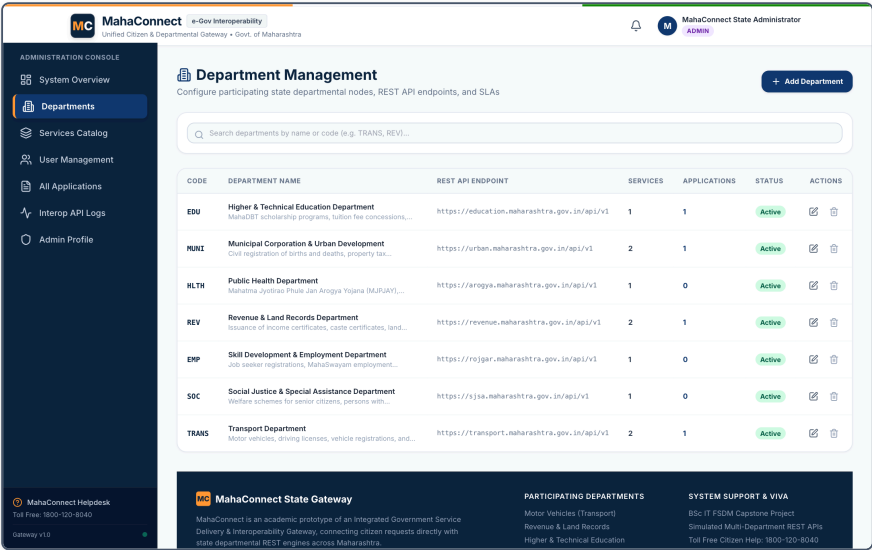


Figure 19: State Department Management Console & Endpoint Configuration

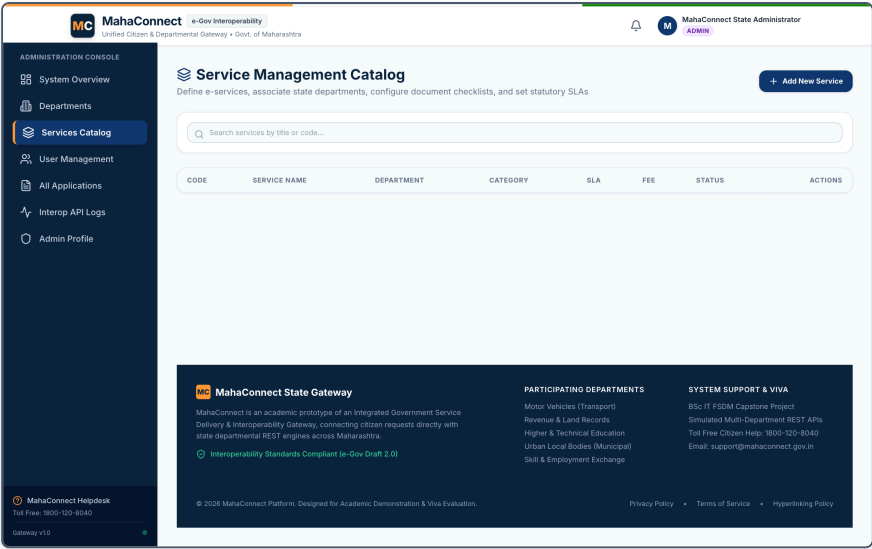


Figure 20: Service Schema Configuration & Dynamic Field Definition

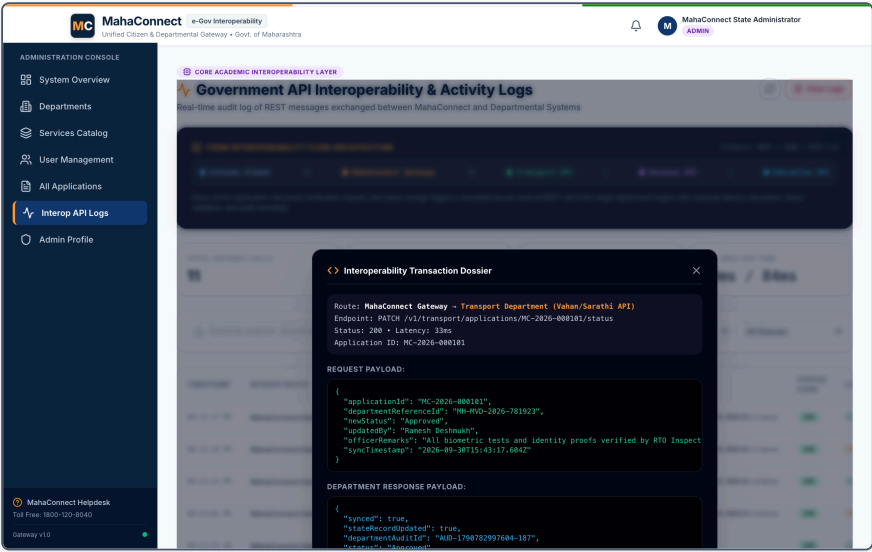


Figure 21: REST API Interoperability Gateway Logs & Modal Payload Inspector

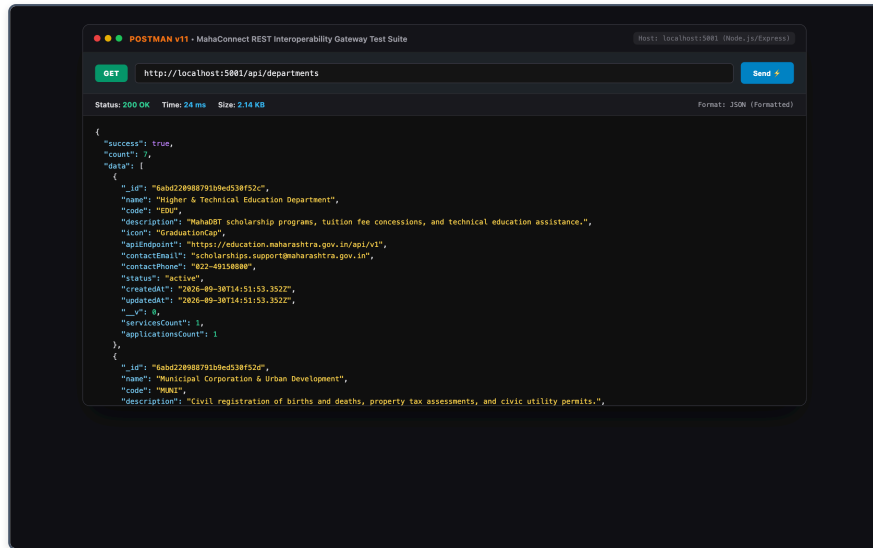


Figure 22: Postman Test Suite: GET /api/departments (Status 200 OK)

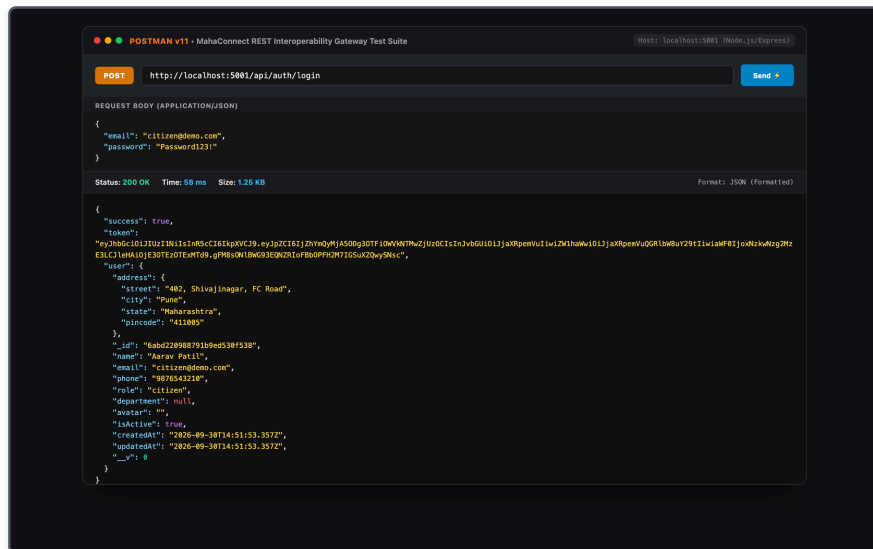


Figure 23: Postman Test Suite: POST /api/auth/login (JWT Token Generation)

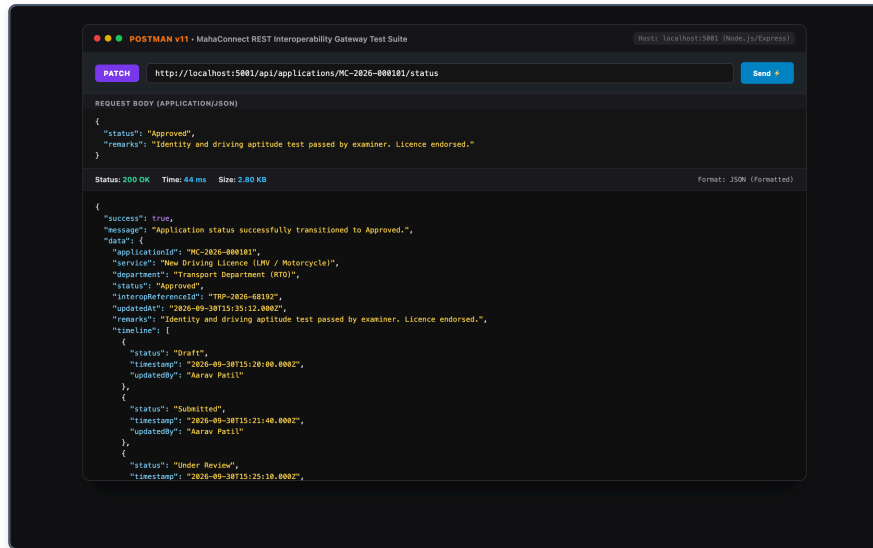


Figure 24: Postman Test Suite: PATCH `/api/applications/:id/status`

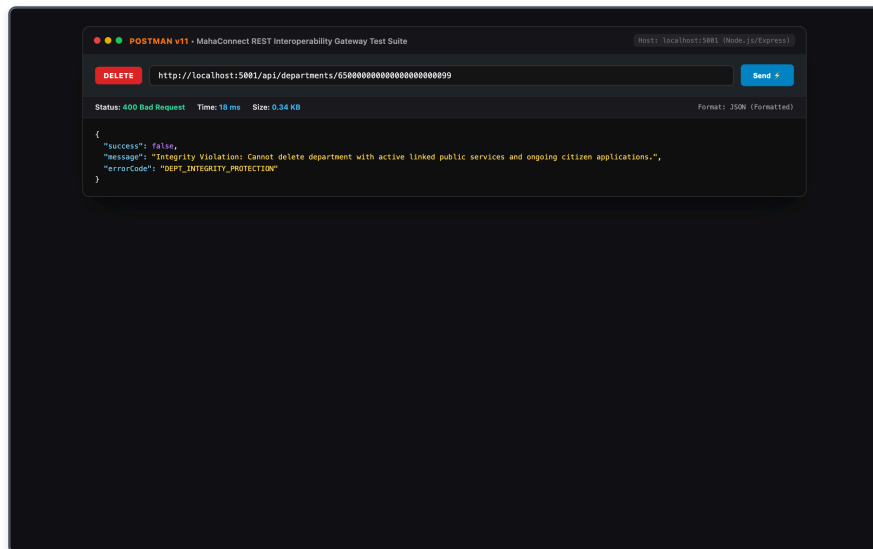


Figure 25: Postman Test Suite: DELETE `/api/departments/:id` (Integrity Check)

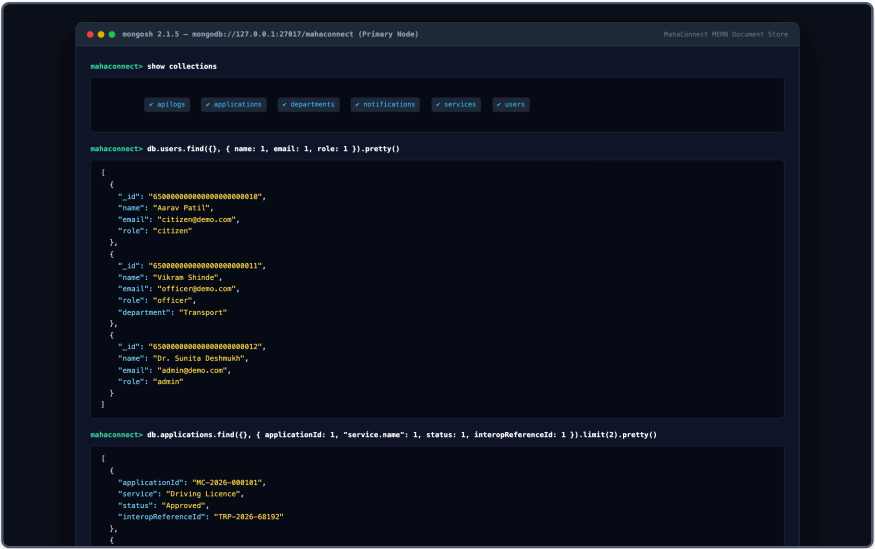


Figure 26: MongoDB Shell (mongosh) Collection & Query Ledger Inspection

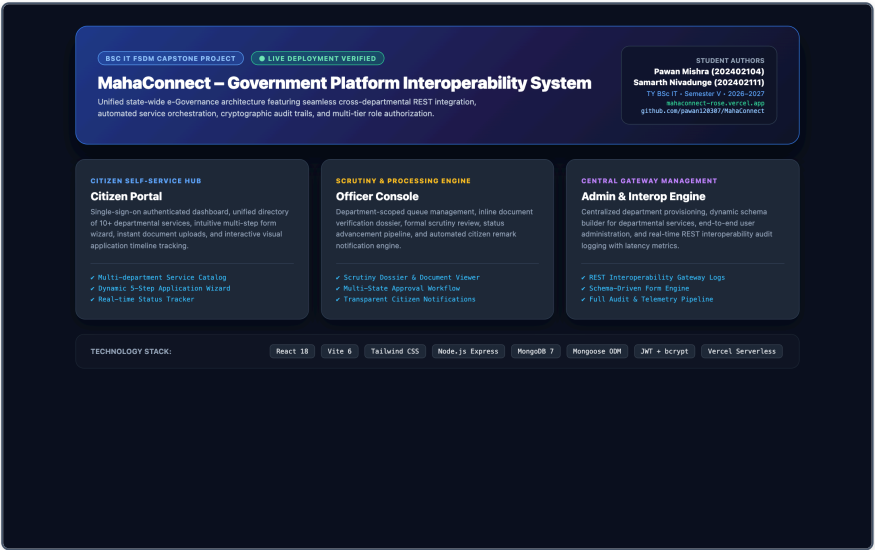


Figure 27: MahaConnect Final Working Multi-Portal Application Showcase

# PROJECT DETAILS

---

|                 |                                                                                       |
|-----------------|---------------------------------------------------------------------------------------|
| DEPARTMENT:     | Information Technology                                                                |
| PROJECT TITLE:  | MahaConnect: Government Platform Interoperability System                              |
| STUDENT NAMES:  | Pawan Mishra<br>Samarth Nivadunge                                                     |
| ROLL NOs:       | 202402104<br>202402111                                                                |
| CLASS:          | TY BSc IT                                                                             |
| SEMESTER:       | V                                                                                     |
| PROJECT GUIDE:  | Mr. Pratharv Surve                                                                    |
| ACADEMIC YEAR:  | 2026-2027                                                                             |
| COLLEGE:        | ZSCT's Thakur Shyamnarayan Degree College                                             |
| DEPLOYMENT URL: | <a href="https://mahaconnect-rose.vercel.app">https://mahaconnect-rose.vercel.app</a> |

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